INCH-POUND

MIL-C-70950 (AR)
AMENDMENT 2
23 January 1992
SUPERSEDING
AMENDMENT 1
24 June 1991

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M148A1B1E1 (M148A2B1)

This amendment forms a part of Military Specification MIL-C-70950 (AR), dated 12 January 1990, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete "M148A1B1E1" and substitute "M148A2B1".

1.1: Delete "M148A1B1E1" and substitute "M148A2B1".

PAGE 2

2.1.2: Delete "M148A1B1E1" and substitute "M148A2B1".

PAGE 5

4.4.1: Delete "M148A1B1E1" and substitute "M148A2B1".

PAGE 6

Table I Title: Delete: "M148A1B1E1" and substitute "M148A2B1".

PAGE 7

Table I Title: Delete: "M148A1B1E1" and substitute "M148A2B1".

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4.5.2.1 Table II: Delete:

"Table II. Sampling

Lot Size	Inspection Level	
	I Major Defects	II Minor Defects
2 to 13	*	*
14 to 90	*	13
91 to 150	125	13
151 to 500	125	30
501 to 1200	125	35
1201 to 3200	125	43
3201 to 10000	125	50
10001 to 35000	296	61

Numbers under inspection levels indicate sample size.

* Indicates one hundred percent inspection.
If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for both inspection levels."

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and substitute:

"Table II Sampling
Sampling Plan for Quality Conformance
(Accept on Zero Defects)

Lot Size	Sample Size (Characteristic)	
	Major	Minor
2 to 8	*	*
9 to 150	32	13
151 to 500	32	30
501 to 1200	74	35
1201 to 3200	74	43
3201 to 10000	87	50
10001 to 35000	109	61

* 100% inspection
** If sample size equals or exceeds lot or batch size, do 100% inspection."

Page 10

4.5.5.1: Delete 4.5.5.1 in its entirety and substitute included 4.5.5.1.

Page 11

4.5.5.2: Delete 4.5.5.2 in its entirety and substitute included 4.5.5.2.

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Page 12

4.5.5.3 Next Higher Assembly: Delete "9323901" and substitute "11735736".

Characteristics 118 and 119, delete "protrusion" and substitute "boss".

PAGE 14

4.5.5.4: Delete 4.5.5.4 in its entirety and substitute included 4.5.5.4.

PAGE 15

Add new page 15a, paragraph 4.5.5.5 included.

Add new page 15b, paragraph 4.5.5.6 included.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.1	Cup	1 OF 1	11735732
			NEXT HIGHER ASSEMBLY: 11735736

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Table II	3.1	Gage
102	Inside diameter	Table II	3.1	Gage
103	Length, total	Table II	3.1	Gage
104	Thickness of lip	Table II	3.1	Gage
105	Taper on diameter	Table II	3.1	Gage
106	Hardness	4.5.6.1	3.3	4.6.1.3
107	Thickness	Table II	3.1	Gage
108	Protective coating missing or inadequate	Table II	3.4	Visual
109	Surface finish	Table II	3.1	Visual
110	True position of I.D.	Table II	3.1	Gage
111	Radius at corner	Table II	3.1	Gage
112	Coating thickness	4.5.6.2.1	3.4	Gage
113	Salt spray	4.5.6.2.2	3.4	4.6.2.1
Minor				
201	Workmanship	Table II	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.5.2	Collar		11735733
			NEXT HIGHER ASSEMBLY
			11735736

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Small outside diameter (boss dia.)	Table II	3.1	Gage
102	Width of boss	Table II	3.1	Gage
103	Depth of boss groove	Table II	3.1	Gage
104	Large outside diameter (flange dia.)	Table II	3.1	Gage
105	Circumferential taper angle of boss	Table II	3.1	Gage
106	Circumferential taper depth of boss	Table II	3.1	Gage
107	True position of large outside diameter	Table II	3.1	Gage
108	Large radius of flange	Table II	3.1	Gage
109	Hardness	4.5.6.1	3.3	4.6.1.2
110	Protective coating missing or inadequate	Table II	3.4	Visual
111	Surface finish	Table II	3.1	Visual
112	Coating thickness	4.5.6.2.1	3.4	Gage
113	Salt spray	4.5.6.2.2	3.4	4.6.2.1
Minor				
201	Small radius of flange	Table II	3.1	Gage
202	Workmanship	Table II	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.4	M148A2B1 Cartridge Case Assembly	1 OF 2	11735736
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness of body material prior to wrapping	Table II	3.1	Gage
102	Total length	Table II	3.1	Gage
103	Distance from flange (base) to first basic diameter	Table II	3.1	Gage
104	Distance from flange (base) to second basic diameter	Table II	3.1	Gage
105	Distance from flange (base) to third basic diameter	Table II	3.1	Gage
106	Distance from flange (base) to fourth basic diameter	Table II	3.1	Gage
107	Inside diameter of mouth	Table II	3.1	Gage
108	True position of mouth I.D.	Table II	3.1	Gage
109	O. D. of mouth	Table II	3.1	Gage
110	Improper number of wraps at mouth	Table II	3.1	Visual
111	Protective coating missing or inadequate	Table II	3.4	Visual
112	Location of weld to edge of inner wrap	Table II	3.1	Gage
113	Location of weld to edge of outer wrap	Table II	3.1	Gage
114	Hardness of body blank	4.5.6.1	3.3	4.6.1.4
115	Assembly tightness	100%	3.5	4.6.5

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.4	M148A2B1 Cartridge Case Assembly	2 OF 2	11735736
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	(cont'd)			
116	Metal defects	100%	3.8.1	4.6.3
117	Salt spray (test panel)	4.5.6.2.2	3.4	4.6.2
118	Salt spray (assembly)	4.5.6.2.2	3.4	4.6.2
119	Distance from base O.D. to inner and outer plies	4.5.6.3	3.1	4.6.6
120	Profile and alignment	100%	3.6	Gage
121	Ballistics	4.5.6.4.2	3.7	4.6.7.2
Minor				
201	Stamping incorrect or illegible	Table II	3.1	Visual
202	45° overlap of body wraps at mouth	Table II	3.1	Gage
203	Workmanship	Table II	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.5.5.5	TITLE Cartons, Ammo, for Interplant Shipment of Steel Cartridge Cases, Empty 37mm to 120mm Inclusive (before sealing)	SHEET 1 OF 1	DRAWING NUMBER 7548235
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Improper assembly	Table II	3.1	Visual
202	Incorrect number of cartridge cases in carton	Table II	3.1	Visual
203	Separator missing or damaged	Table II	3.1	Visual
204	Workmanship	Table II	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.5.5.6	TITLE Cartons, Ammo, for Interplant Shipment of Steel Cartridge Cases, Empty 37mm to 120mm Inclusive (after sealing)	SHEET 1 OF 1	DRAWING NUMBER 7548235
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Carton damaged, cut or gouged through all layers	Table II	3.1	Visual
202	Carton improperly sealed	Table II	3.1	Visual
203	Contents loose	Table II	3.1	Visual/Manual
204	Marking missing, incorrect or illegible	Table II	3.1	Visual
205	Workmanship	Table II	3.8	Visual

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4.5.6.2.2: Delete: "Five samples shall be randomly selected for testing." and substitute:

"Five samples of each cartridge case component, five 4" x 6" test panel samples with a minimum of one sample per coil, and five cartridge case assemblies shall be randomly selected for testing."

PAGE 17

* 4.5.6.4.1 Table III:

Item 5: Delete: "Propellant gas leakage (Cup burn mark)" and substitute: "Propellant gas leakage between Body and Cup (Cup burn mark extending more than 1/4" aft of the joint)".

Item 6: Delete: "Propellant gas leakage (Body burn mark) 3, 4".

Item 7: Delete: "7" and substitute "6".

Add: "7. Crack or split through Cup 0 1
8. Crack or split through Base 0 1".

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* 4.5.6.4.2 Table IV:

Item 5: Delete: "Propellant gas leakage (Cup burn mark)" and substitute: "Propellant gas leakage between Body and Cup (Cup burn mark extending more than 1/4" aft of the joint)".

Item 6: Delete: "Propellant gas leakage (Body burn mark) 2, 3 Not permitted".

Item 7: Delete: "7" and substitute "6".

Add: "7. Crack or split through Cup 0 1 Not permitted
8. Crack or split through Base 0 1 Not permitted".

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* 4.6.7.1.4.4 Delete:

"The cartridge case shall be examined for entry of propellant gas into the gun chamber by passing through any component or component interface. Mechanisms of failure include but are not limited to burn-through, rupture, joint failure, cracks and gas flow between body plies. Leakage is evidenced by the presence of burn marks on the component causing failure."

and substitute:

"The cartridge case shall be examined for leakage of propellant gas into the gun chamber to the rear of the Body-Cup interface. Mechanisms of failure include but are not limited to burn-through, rupture, joint failure, cracks and splits. Leakage is evidenced by the presence of burn marks on the Cup or Base."

The margins of this Amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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